Statistics – Scenario-Based Questions | Set 5

Scenario 1: One-Way ANOVA for Regional Sales

Question: A company wants to analyze whether there is a significant difference in the average sales across four regions using One-Way ANOVA.

Answer: First, I would group the sales figures by North, South, East and West and calculate the average for each region. Then I would use a one-way ANOVA to check whether the regional means differ more than we would expect from normal variation. The null hypothesis is that all four regions have the same average sales. If the p-value is below 0.05, I would conclude that at least one region has a different average and follow up with a post-hoc test to find out which ones. I would also look at a box plot and check the ANOVA assumptions before making a decision.

Scenario 2: Two-Way ANOVA for Employee Performance

Question: A company wants to analyze whether the employee performance scores are affected by the type of training program attended and the employee's experience level. How can Data Science concepts be applied using a Two-Way ANOVA?

Answer: I would compare performance scores based on both training type and experience level. A two-way ANOVA helps check three things: whether training matters, whether experience matters, and whether the effect of training changes with experience. I would first clean the data, calculate group averages and draw a comparison chart. After checking the model assumptions, I would look at the p-values for both factors and their interaction. This would help the company understand which training programs work better for different employees.

Scenario 3: VIF for House Price Prediction

Question: A real estate company wants to build a model to predict house prices using features like area, number of bedrooms, number of bathrooms, and overall house size. How can Data Science concepts be applied to check for multicollinearity using VIF?

Answer: I would start by checking whether the input features carry overlapping information. For example, area and overall house size may describe almost the same thing. I would calculate the Variance Inflation Factor (VIF) for each predictor, using a constant in the regression setup. A high VIF, often above 5 or 10, suggests strong multicollinearity. I would review the correlated variables and consider keeping the most useful one rather than removing features automatically. Then I would rebuild the model and check whether its predictions and coefficient stability improve.

Scenario 4: Independent T-Test for Teaching Methods

Question: A school tested two teaching methods: one traditional and one interactive. After the course, both groups took the same exam. The school wants to know if there’s a significant difference in their average scores.

Answer: I would separate the exam scores into traditional and interactive groups and compare their averages. An independent two-sample t-test can show whether the observed difference is statistically significant. The null hypothesis says the average scores are equal. I would check the score distributions and use Welch’s t-test if the group variances are unequal. If the p-value is below 0.05, I would reject the null hypothesis. I would also report the mean difference and sample size, because a higher average alone does not prove that one method is better.

Scenario 5: Pearson Correlation for Workout Time

Question: A fitness center wants to understand if increasing workout time leads to more calorie burn. How can data science concepts be applied to analyze this relationship?

Answer: I would collect workout duration and calories burned for each session and start with a scatter plot. Then I would calculate Pearson's correlation coefficient to measure the strength and direction of the linear relationship. A positive value close to 1 would suggest that longer workouts are linked with higher calorie burn. I would also check the p-value and look for unusual data points. Correlation alone cannot prove that extra workout time causes the increase, since workout intensity and other factors can also affect calories burned.